

Command Diagnosis — Cristopher Sánchez (LHP)

Is he near the zone? · 2021–2026 · data through 2026-09-15 · prepared 2026-09-16

Prepared for: pitching coach · pitching analyst · catcher · manager · Sánchez

Throws: L · **Arsenal:** sinker, changeup, slider (every 2026 pitch)

Governance: Use Case #45 (uc-pps-031) · build dp_uc45 · Baseball Functions kernel inherited verbatim · governed **Edge Rate** (UC8) inherited verbatim · **NEW, provisional:** Shadow Zone family SZ-0...SZ-5, CL-1 count_Leverage (PD-7 promoted), PN-1 pitcher peer delta, ZC-1 common-rail re-score · new data finding **O-18**

⚠ **Read this first: data window, sample sizes, and a rulebook change.** • Source: phils_2021...2026.parquet , regular season, deduplicated; entity lock pitcher == 650911 . Cache current through **2026-09-15** (Sánchez started that night). 31 starts in 2026, **2,960 pitches / 805 PA**. • **2021 and 2022 are combined** (873 pitches, a swingman sample). Read that column loosely. • 2026 is split at the **All-Star break** (first half through 07-13: **1,903 pitches / 525 PA**; second half: **1,057 pitches / 280 PA**, 11 starts). • **The 2026 strike zone is defined differently from 2025 (O-18)**. Every hitter now has one fixed zone top and bottom (the height-based ABS zone), and for the same hitters the top of the zone sits a median **0.23 ft (2.8 in) lower** than in 2025. Every pitcher's zone numbers moved. Section 3 separates Sánchez from the rulebook, so don't read a 2025→2026 zone number before you've read it. • Walk rate is the kernel's bbrate (unintentional BB / PA). Sánchez has **0** intentional walks since 2021, so here it equals public BB%.

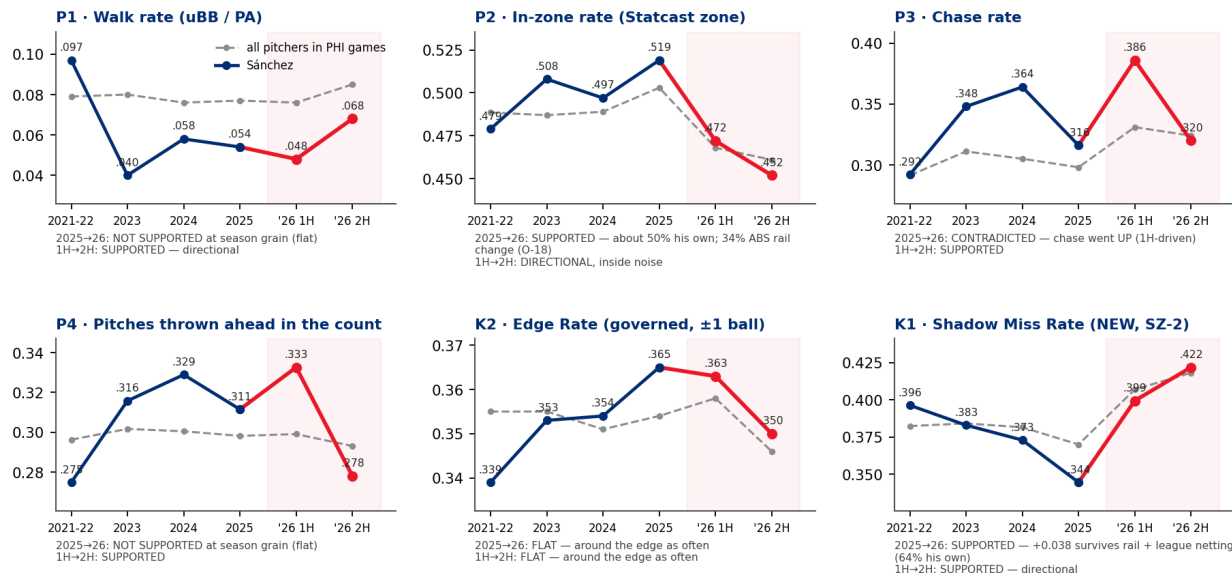
Bottom line

1. **At the season level, only one of your four premises holds up.** His walk rate is flat (**.054 → .055**), chase went **up** (.316 → .362) and so did his share of pitches thrown ahead (.311 → .313, essentially flat). His in-zone rate did drop (**.519 → .465**), but about a third of that drop is the new zone definition.
 2. **Since the All-Star break, all four premises hold.** Walks **.048 → .068**, in-zone **.472 → .452**, chase **.386 → .320** ($p = .008$), pitches thrown ahead **.333 → .278** ($p = .002$). League numbers in Phillies games barely moved over the same stretch.
 3. **He's around the edge as often as ever, but he misses the shadow more often.** The governed Edge Rate is flat (**.365 → .358**). His Shadow Miss Rate (share of pitches more than one baseball outside the zone) rose from **.344 to .407**. Put 2025 on 2026 rails and net out the league, and **+.038 of that is his own (64%)**. In 2025 he beat the median starter by a wide margin. In 2026 he is the median starter.
 4. **The proof: hitters stopped chasing his misses.** On pitches outside the shadow, first-half hitters swung **33.3%** of the time. That spike produced the largest 2025→2026 rise in chase beyond the shadow among 11 comparable arms. Since the break it's **26.0%** ($z = -2.65$, $p = .008$), while the league held at .274 → .276. When he was ahead in the count, the rate fell from **.401 to .307**, which is league average (.320).
 5. **The leak is the changeup to right-handed hitters.** In the second half, **62.1%** of the changeups he throws to righties miss the shadow (up from 54.1%), and righties chase those misses **30.2%** of the time, down from 39.9%. **Put plainly: he is missing the shadow only slightly more often than in the first half, but the misses have moved to spots hitters don't chase.** The fix is getting the changeup back into the shadow, not throwing more strikes in the heart of the zone.
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1 · The premises, adjudicated

Cristopher Sánchez — the four premises, the governed Edge Rate and the new Shadow Miss Rate

Seasons 2021-22 → 2025, then 2026 split at the All-Star break (red). Dashed = every pitcher in a Phillies game that season. 2026 zone rails changed definition (O-18) — see Fig. 4 before reading 2025→2026 zone moves.



#	Premise	2025 → 2026 season	2026 1H → 2H	Verdict
P1	Walk rate is up	.054 → .055 (44 BB each, 809 / 805 PA), p = .98	.048 → .068 (25/525 → 19/280), p = .23	Second half only , and directional. Add HBP and it's .051 → .082
P2	In the zone less	.519 → .465, p < .001; league .503 → .465	.472 → .452, p = .29; league .468 → .461	Yes, but only half is his. On common rails, 50% of the geometric drop is his own and 34% is the rail change (§3)
P3	Less chase	.316 → .362, up, p = .008	.386 → .320, p = .008; league .331 → .324	Season: contradicted. A first-half chase spike hid the problem. Second half: supported
P4	Ahead in the count less	.311 → .313, p = .89	.333 → .278, p = .002; league .299 → .293	Second half only , and strongly. Pitches thrown behind rose from .229 to .273, and PAs reaching three balls from .149 to .196

Where "ahead" comes from. This is the existing count-leverage definition PD-7 (`uc-pps-017`), now promoted to a function (CL-1): the share of pitches thrown with fewer balls than strikes before the pitch. First-pitch strike rate tells the same story from the other end: **.661 → .629** across the break.

Were results actually worse? A little. Kernel wOBA allowed rose from **.263 (2025) to .298 (2026)**, but expected wOBA only moved from **.279 to .286**, and his strikeout rate went *up* (.262 → .267). The second half is where the expected numbers moved too: xwOBA **.279 → .299**, K% **.274 → .254**.

2 · Edge vs shadow: two different questions, one geometry

You asked for two things. First, the edge rate you had already defined: pitches within one baseball of the zone line, inside or outside. Second, a new "shadow" measure: a target one baseball larger than the zone, and how often he misses it completely.

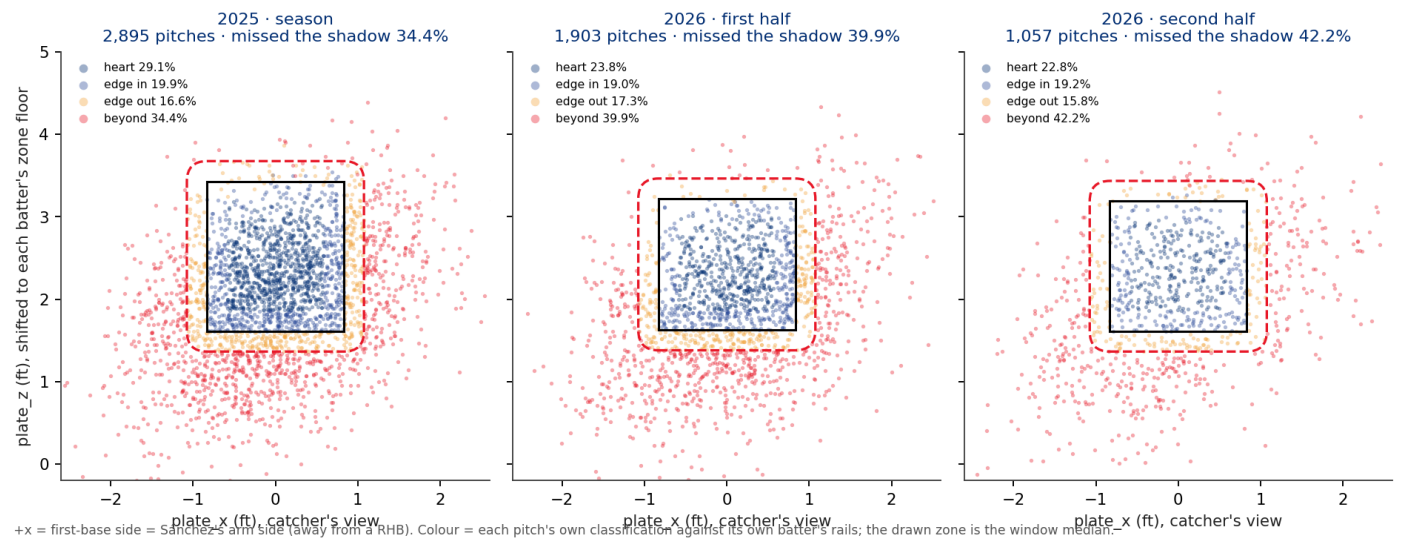
The edge rate already exists and is governed. `edge_rate(level, df)` dates to UC8, has glossary approval, carries disposition **A** in the Baseball Functions intake register, and uses the ratified constant `BALL_FT = 2.94/12 ft`. It is carried here verbatim. *It has not yet been pasted into `Baseball Functions.ipynb` (escalation E-2).*

The shadow is built on the edge's geometry, so the two can't drift apart. SZ-0 classifies every tracked pitch with the same distance function as `edge_rate`:

Region	Definition	2025	2026 1H	2026 2H
heart	inside, > 1 ball from every edge	.291	.238	.228
edge in	inside, ≤ 1 ball from an edge	.199	.190	.192
edge out	outside, ≤ 1 ball from the perimeter	.166	.173	.158
beyond	> 1 ball outside: "missed the shadow"	.344	.399	.422

By construction, **Edge Rate** = edge in + edge out, **Shadow Zone Rate (SZ-1)** = heart + edge in + edge out, and **Shadow Miss Rate (SZ-2)** = beyond. The build asserts that SZ's edge twin matches the governed `edge_rate` in every season (max difference 0.0004, rounding only).

The shadow: rulebook zone (black) grown by one baseball, 2.94 in (red dashed, rounded corners)



Semantic consistency with other definitions in the repo. Two other "shadow" definitions already exist. The organization reconciled against both rather than adding a fourth (`out/dp_uc45_geometry_crosswalk.csv`):

- **OZ family** (`uc-pos-005`, batter side) uses a *rectangular* ±1-ball band. It disagrees with SZ only at the four corners: **38 of 10,985** Sánchez pitches (0.35%).
- **Attack Zone** (`uc-pps-022`) calls anything within **0.33 ft** outside the zone "shadow". That constant comes from UC #11 and was still being used on 07-25, the day after the ball-width ruling, which it doesn't match. It would move **5.5%** of Sánchez's pitches from "beyond" back into the shadow. Escalated as E-3; SZ does not use it.
- The geometric zone (`plate_x`, per-pitch rails) and Statcast's `zone` attribute disagree on **4.0%** of pitches. The kernel's `in_zone_rate` (which uses `zone`) is still the governed number for P2; the geometric twin is used only where a re-score needs geometry.

3 · 2026 changed the zone (O-18) — how much of the move is Sánchez?

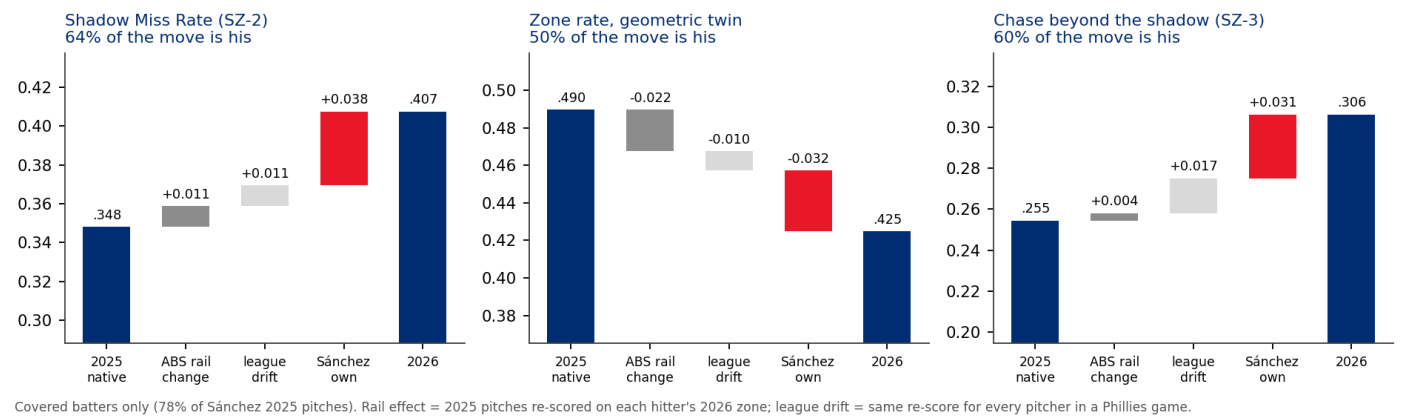
The 2026 cache defines the zone differently. For batters with 100+ pitches, **100% of 2026 batters have a single fixed zone top and bottom** (within-batter SD 0.000 ft, vs 0.071 in 2025). For the **32** hitters seen 100+ times in both seasons, the top of the zone fell a median **0.234 ft**, and it fell for **84%** of them. The bottom barely moved (+0.008). That matches MLB's published ABS zone: top at 53.5% of the batter's height, bottom at 27%. The external definition is used here only to explain what the data shows; it enters no computation.

Every pitcher's zone numbers moved with it. Across all pitchers in Phillies games, in-zone rate fell **.503 → .465** and Shadow Miss Rate rose **.370 → .411**.

Two controls, and why the second one is the one to trust.

- **PN-1 (peer-netted, native rails)** compares Sánchez with the 12 arms in the log that threw 200+ pitches in both seasons. It says his shadow-miss rise (+.063) matches the peer median (+.060). **This control is misleading here:** how much the lower zone top hurts a pitcher depends on where he locates vertically, and Sánchez lives low.
- **ZC-1 (common-rail re-score)** re-scores every 2025 pitch against the same hitter's 2026 zone (78% coverage). Measured that way, the rail change moves his 2025 Shadow Miss Rate only from **.348 to .359**. The league re-scored the same way moves only **+.011** from 2025 to 2026.

2026 changed the zone, not just the pitcher: the move decomposed on common (2026 ABS) rails



Metric (covered batters)	2025 native	Rail change	League drift	Sánchez's own	2026
Shadow Miss Rate	.348	+.011	+.011	+.038 (64%)	.407
Zone rate, geometric	.490	-.022	-.010	-.032 (50%)	.425
Chase beyond the shadow	.255	+.004	+.017	+.031 (60%)	.306
Edge Rate	.351	+.010	-.005	+.003	.358

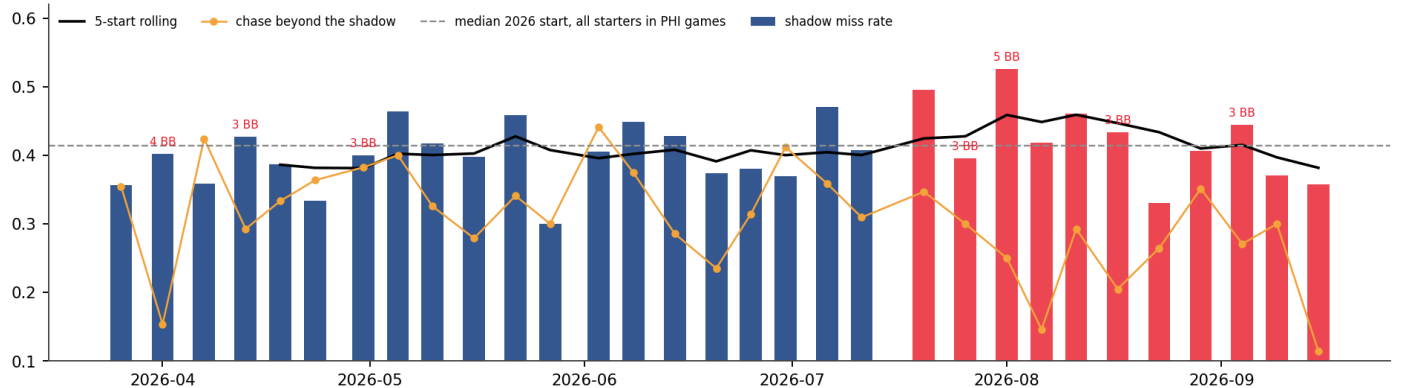
Re-run on common rails, PN-1 agrees with ZC-1. Against the 11 covered arms, Sánchez's Shadow Miss Rate rose **+.027 more than the peer median** (4th-largest rise), his zone rate fell **-.031 more** (3rd-largest drop), and his chase beyond the shadow rose **+.039 more, the largest rise in the group**.

4 · So is he near the zone?

He's near the edge as often as ever, and missing the shadow more. Both of the outcomes you described are happening, in sequence:

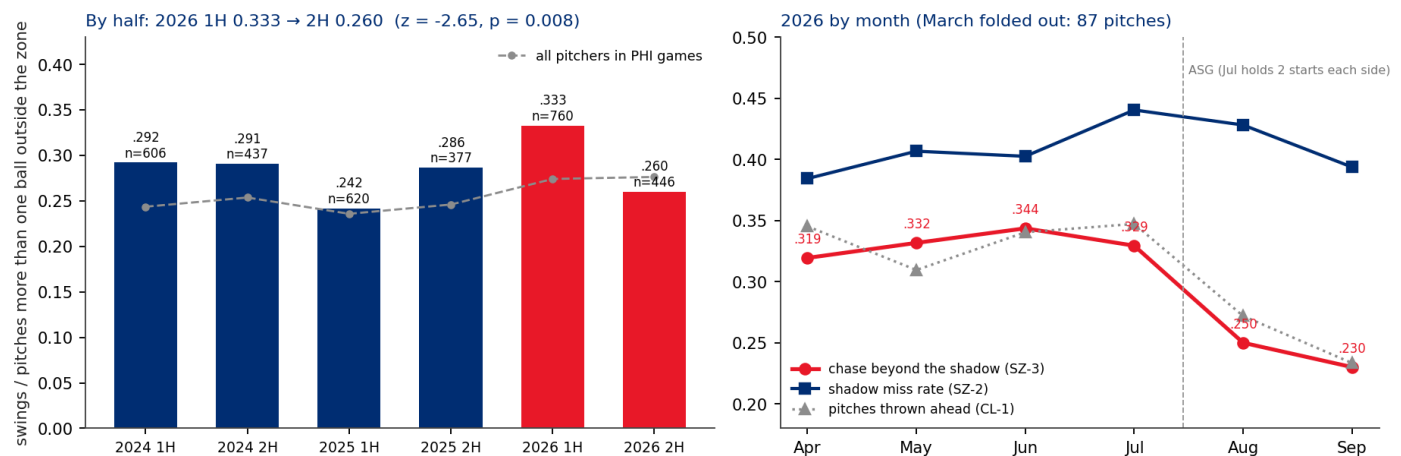
- **All season, he has missed the shadow more often.** This is real and not just the rulebook: +.038 of his own. In 2025 his median start missed the shadow **.371** of the time in the first half and **.311** in the second, against a field median of about **.37**. In 2026 his median start is **.401 / .418** against a field median of **.413 / .415**. He went from elite to average at this.
- **Since the break, the misses land farther outside and in different spots.** The median miss lands **0.659 ft** beyond the zone perimeter, up from 0.595 in the first half. Low misses fell from **55.9%** of misses to **43.5%**. Glove-side misses (in on a right-handed hitter's hands) rose from **14.9% to 23.8%**, and high misses from 2.9% to 5.4%. A low miss gets chased; a miss in on a righty's hands or up usually doesn't (`dp_uc45_miss_direction.csv`).

Every 2026 start (31): share of located pitches that missed the shadow; red = after the break



5 . The proof — chase when the pitch is outside the shadow

The proof: since the break, hitters stopped chasing the pitches that miss the shadow



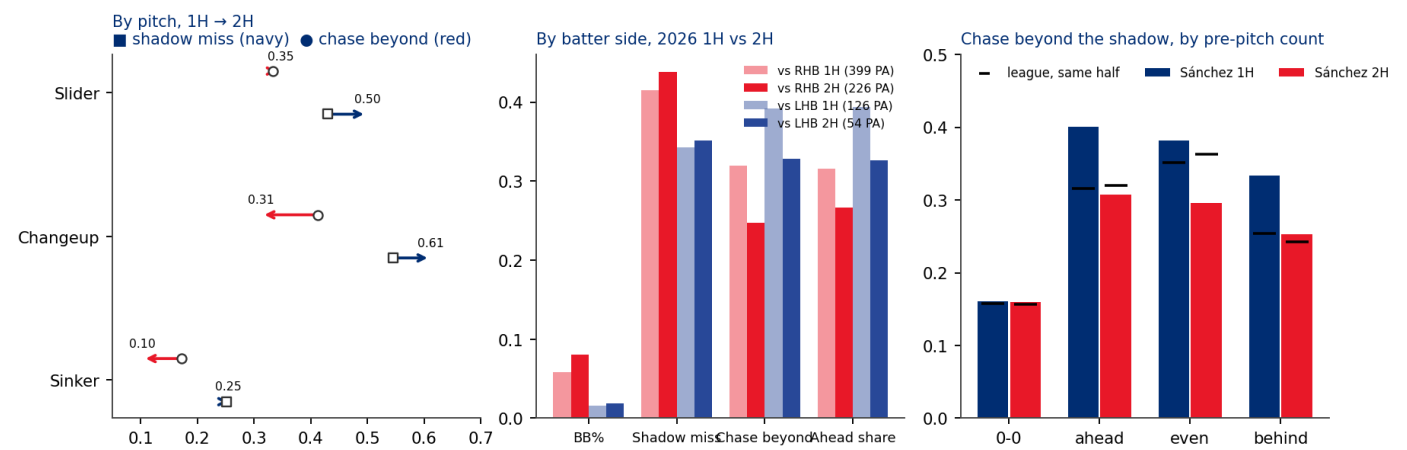
2026 split	Beyond-shadow pitches 1H / 2H	Chase 1H → 2H	p	League, same split
All pitches	760 / 446	.333 → .260	.008	.274 → .276
vs RHB	617 / 376	.319 → .247	.016	.279 → .273
vs LHB	143 / 70	.392 → .329	.37	.270 → .280
Changeup	407 / 218	.413 → .312	.013	.319 → .305
Sinker	203 / 127	.172 → .102	.08	.216 → .200
Slider	150 / 101	.333 → .347	.83	.311 → .302
Pitcher ahead in count	322 / 166	.401 → .307	.043	.315 → .320
Two strikes	284 / 148	.416 → .378	.46	.365 → .384

How to read this. In the first half, hitters chased his misses far more than the league did: .333 against .274, and .401 against .315 when he was ahead. That was his highest half since 2023, and it covered for a shadow-miss rate that had already risen. **In the second half that chase fell back to league level.** He kept missing the shadow as often (.399 → .422, p = .23), but those misses stopped turning into free strikes. Counts went the other way: pitches thrown ahead .333 → .278, three-ball PAs .149 → .196, walks .048 → .068.

The month view says the same thing: chase beyond the shadow ran **.319 / .332 / .344 / .329** from April through July, then fell to **.250** in August and **.230** in September.

6 · Where it leaks

Where the free swings went: right-handed hitters, the changeup, and counts he was ahead in



- **Right-handed hitters are the problem.** Against RHB in the second half: BB% **.080** (226 PA), Shadow Miss Rate **.438**, chase beyond the shadow **.247**, and pitches thrown ahead **.267**. Against LHB the season Shadow Miss Rate is flat (.342 → .346) and so is the walk rate (.017 in 2026).
- **The changeup accounts for most of it.** Against RHB its Shadow Miss Rate went **.541 → .621** (695 → 330 pitches) and its beyond-shadow chase **.399 → .302**. The pitch still misses bats when hitters swing (whiff **.438 → .465**), so the problem is where it ends up, not the pitch itself.
- **The sinker is fine.** Its Shadow Miss Rate barely moved against RHB (**.270 → .256**) and it still lands in the zone **.597** of the time. It was never getting chased much (.127 → .080).
- **The slider misses more but still gets chased.** Shadow Miss Rate **.430 → .500**, chase **.333 → .347**. Watch it; it's not the cause.

7 · Your notebook cell, reproduced

The organization ran your cell as written: kernel functions, month grain for 2026, season grain for 2023–25 (out/ dp_uc45_notebook_reproduction.csv).

game_year	month	pitches	PA	bbrate	in_zone	chase	barrel	hard hit	FPSR
2023	All	1460	396	.040	.508	.348	.082	.402	.665
2024	All	2797	755	.058	.497	.364	.054	.345	.658
2025	All	2896	809	.054	.519	.316	.057	.398	.653
2026	4	562	159	.082	.475	.359	.104	.415	.692
2026	5	504	143	.021	.478	.388	.053	.436	.671
2026	6	564	152	.046	.465	.417	.078	.408	.645
2026	7	386	103	.049	.451	.377	.067	.427	.631
2026	8	570	148	.074	.451	.313	.033	.367	.615
2026	9	287	79	.063	.456	.308	.100	.350	.620

March (1 start, 87 pitches) is in the receipt and left out here. **Known-defect exposure in this cell:** `whiff_rate` 's inner join (D-1/D-2) would drop **2** of 135 appearances (2021–26) at game grain but drops nothing at month grain. `fpsr` returns only groups that threw at least one first-pitch ball, which drops **1** appearance at game grain. `chase_rate` counts a null `zone` as in-zone (D-7): **1** pitch in 2025. `hard_hit_rate` counts untracked balls in play as not hard-hit (O-8): **8** balls in play. None of these changes a published number.

Actions

Pitching coach / Sánchez. The target is the shadow, not the heart of the zone. Adding strikes over the middle would give up damage he isn't currently allowing (xwOBA on contact is only .361 → .365 across the break). **Get the changeup to righties back within one baseball of the zone:** its misses have drifted from low and chaseable to far outside and easy to take. Measure progress by Shadow Miss Rate on that pitch against RHB (2026 2H vs RHB **.621**; 2025, both sides, **.490**).

Catcher. When he's ahead, righties have stopped chasing (.401 → .307). Set the target *inside* the shadow on the chase pitch; a target at the bottom edge is being missed by more than a ball.

Analyst. Any 2025→2026 zone comparison should go through ZC-1 (common rails) first. Peer-netting on native rails hides the fact that low-zone pitchers were affected less by the lower zone top (O-18). Tripwires are armed in 07 §4.

Manager. Results have moved less than process: season xwOBA is .279 → .286. Treat this as a location problem with one pitch against one side of the plate, not a general decline.

Candid data-window & freshness caveats

- **Sample sizes.** The second half is **11 starts, 280 PA**. Walk-rate and in-zone changes across the break are *directional* (p = .23 and .29). The chase and count-leverage changes are the ones that clear significance.
- **O-18 is a definition change, not a data error.** ZC-1 covers only batters who appear in 2026 (78% of Sánchez's 2025 pitches). The Statcast `zone` attribute can't be re-scored, so P2's decomposition uses the geometric twin, which disagrees with `zone` on 4.0% of pitches.
- **The PN-1 cohort is thin** (12 arms native, 11 on common rails) and mixes Phillies staff with opponents who pitched a lot against the Phillies. It is a Phillies-schedule cohort, not a league cohort.
- **League controls are every pitcher who pitched in a Phillies game**, not MLB as a whole.
- **Not examined here:** release point and arm angle (`arm_angle` is only 77.7% complete for Sánchez in 2026 — DQ-17 WARN), catcher effects (a battery split needs the catcher × game receipt), ABS challenge outcomes (not in the cache), and pitch-shape changes.
- **wOBA method.** Kernel wOBA (FanGraphs weights over PA) sits about .01 below Statcast's `woba_value` scale, as documented in `uc-pps-019` §05. The kernel number is the governed one.

- **The 20-80 scouting grades (SG-1) were considered and not used.** Under a single zone definition, the pitcher-season population in the log is only 13 arms at 500 pitches — too few to grade against.

Receipts: `out/dp_uc45_*.csv` (29) · figures `out/dp_uc45_fig1...fig6` · interactive: `dp_uc45_command_dashboard.html` · governance: `00 – 07` in this folder.